

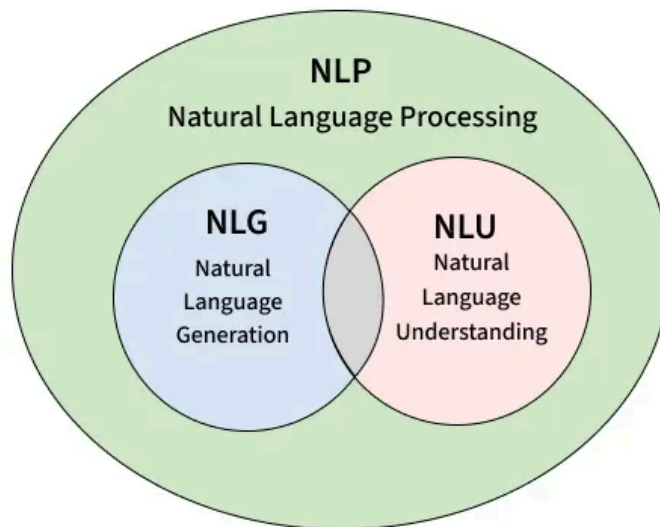
Artificial Intelligence | Natural Language Generation

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Natural Language Generation (NLG) is a subfield of Artificial Intelligence and Natural Language Processing (NLP) concerned with automatically generating contextually relevant text from structured or unstructured data. It enables machines to articulate thoughts or observations in natural language.

While NLP covers the spectrum of interpreting and producing language, NLG specifically focuses on the production aspect like turning raw data into readable text. It powers applications ranging from automated news articles and personalized emails to financial reporting systems and intelligent chatbots.



NLP vs NLG vs NLU

NLG is the task of generating text that resembles human writing, given a structured or semi-structured input. This input can be in the form of tabular data, knowledge graphs or programmatic events. NLG can be viewed as the reverse of [Natural Language Understanding \(NLU\)](#). While NLU transforms human language into machine-readable formats (parsing questions or commands), NLG transforms machine-represented information into human-friendly text.

How does NLG work

A typical NLG pipeline consists of the following stages:

1. **Content Determination:** The system decides which information from the input data is relevant and should be mentioned. This involves filtering facts based on context or importance.
2. **Document Structuring:** The content is organized into a continuous structure. Decisions are made about the order in which topics or facts should be presented.
3. **Aggregation:** Facts are grouped to improve fluency and reduce redundancy. This ensures the text reads naturally, like how a human would summarize multiple data points.

- 4. **Lexicalization:** Appropriate words and expressions are chosen to represent the facts.
- 5. **Referring Expression Generation:** The system generates references to entities such as “it”, “they” or proper names to maintain clarity and consistency across the text.
- 6. **Linguistic Realization:** Grammar rules are applied to construct well-formed sentences.

Techniques for Evaluating NLG Systems

Evaluating the effectiveness of generated text is a complex but critical task. Common evaluation techniques include:

- 1. **Task-Based Evaluation:** Measuring whether the text helps humans perform a specific task (e.g., decision-making from medical summaries).
- 2. **Human Judgment:** Evaluators rate text quality based on fluency, coherence and informativeness.
- 3. **Automated Metrics:** BLUE, ROUGE and METEOR compare generated outputs with reference texts written by humans, often used in summarization and translation tasks.

Differences between NLP, NLG and NLU

[Natural Language Processing \(NLP\)](#), Natural Language Generation (NLG) and Natural Language Understanding (NLU) are three distinct but linked areas of natural language processing. Here's a brief overview of the differences between them:

Aspect	NLP	NLG	NLU
Input	Raw or structured language	Structured data	Natural language text
Output	Structured or unstructured text	Human-readable text	Machine-readable meaning
Goal	Interpret and produce language	Generate natural-sounding text	Understand meaning and intent
Techniques Used	Parsing, tagging, vectorization	Templates, ML models, transformers	Syntax analysis, semantics, embeddings
Tasks	Translation, speech-to-text, summarization	Report writing, product descriptions	Intent detection, sentiment analysis
Common Tools	spaCy, NLTK, Hugging Face	GPT, T5, SimpleNLG	BERT, RoBERTa, Dialogflow
Evaluation Metrics	Accuracy, F1-score	BLEU, ROUGE	Precision, recall, intent accuracy

Applications of Natural Language Generation

Natural Language Generation has seen increasing usage across domains where large volumes of structured data need to be communicated in a human-readable form:

1. **Automated Reporting:** Used in journalism, finance and weather forecasting.
2. **E-Commerce and Product Descriptions:** NLG systems generate descriptions for thousands of products using specifications from databases, saving manual writing effort and ensuring consistency.
3. **Customer Service & Chatbots:** Virtual assistants and bots use NLG to craft responses based on user intent and backend system data.
4. **Medical Reporting:** Patient summaries and diagnosis is generated from structured Electronic Health Records (EHRs), reducing documentation workload for clinicians.
5. **Business Intelligence Dashboards:** Data dashboards can narrate insights using natural language, improving accessibility for non-technical stakeholders.

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